AI Fundamentals: Language and Vision in AI

Understanding artificial intelligence, natural language processing, and computer vision.

ABINRAJ CHOVULLAKANDY · IBM SKILLSBUILD · 20 OCTOBER 2026



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THE BIG PICTURE

AI turns data into useful action

Artificial intelligence enables computers to perform tasks associated with human intelligence, including learning, language understanding, image recognition, problem solving, and prediction.



Language AI

Processes and generates human language.



Vision AI

Analyzes images and video.



Predictive AI

Estimates future outcomes.



Generative AI

Creates text, images, and code.

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LANGUAGE AI

Language AI helps computers understand people

Natural Language Processing enables systems to process, analyze, and generate written or spoken language.

Tokenization: split text into smaller units.

Classification: organize text into categories.

Sentiment: identify positive, negative, or neutral opinions.

Translation, summarization, and entities: transform text and extract meaning.



Applications include chatbots, spam detection, feedback analysis, and writing assistance.

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A LANGUAGE WORKFLOW

Language AI moves from input to evaluated output

01

Receive

A question, sentence, or document enters the system.

02

Prepare

The text is prepared and patterns are identified.

03

Interpret

A trained model interprets the input.

04

Generate

The system produces a response, translation, summary, or classification.

05

Check

The result is checked for accuracy and usefulness.

Example: “I love this product. It works perfectly!” may be classified as positive. Important outputs still need verification.

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VISION AI

Vision AI helps computers interpret images

Vision AI analyzes visual information to identify objects, recognize patterns, and extract useful information.

- **Classification** labels an entire image.
- **Detection** finds objects and their locations.
- **Segmentation** identifies regions at pixel level.
- **OCR** extracts text from images and scans.



Applications include medical imaging, traffic monitoring, quality inspection, and crop monitoring.

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A VISION WORKFLOW

Vision AI learns from pixels to predictions

01

Receive

An image or video frame enters the system.

02

Adjust

The input is resized or adjusted.

03

Learn

The model learns shapes, edges, colors, and textures.

04

Predict

It predicts objects, labels, or features.

05

Interpret

The output is interpreted for a task or decision.

Manufacturing example

Cameras can flag possible scratches, cracks, or defects for further inspection.

Accuracy has conditions

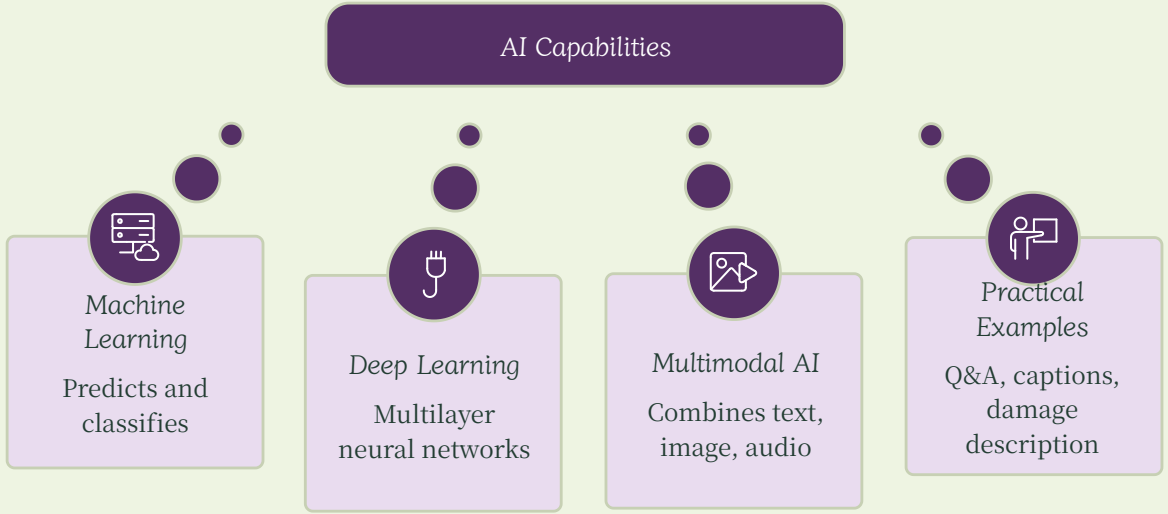
Blurry images, poor lighting, or unfamiliar conditions can reduce accuracy.

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HOW CAPABILITIES CONNECT

ML, deep learning, and multimodal AI work together

Machine learning learns patterns from data for predictions or classifications. Deep learning uses multilayer neural networks to learn complex patterns.

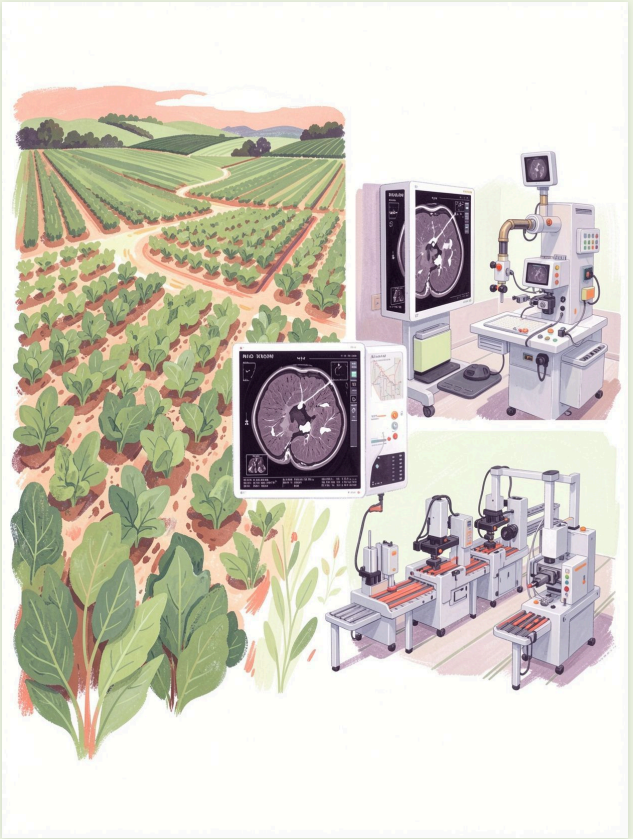


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FROM CONCEPT TO PRACTICE

AI is changing work across industries

- **Education:** language learning, writing support, tutoring, summarization.
- **Healthcare:** medical-image support and clinical-text organization.
- **Business:** chatbots, email classification, feedback analysis, document processing.
- **Manufacturing and transport:** defect detection, quality inspection, traffic monitoring.
- **Agriculture:** crop monitoring, plant-disease signs, weed detection.



Benefits include faster processing, automation, productivity, accessibility, and decision support.

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RESPONSIBLE USE

Responsible AI keeps people in the loop

Data quality

Incomplete or inaccurate data can reduce performance.

Fairness

Evaluate differences in performance across groups.

Privacy and security

Protect sensitive information and guard against misuse.

Transparency and accountability

Explain use, limits, and responsibility.

Reliability and oversight

Test continuously and keep people involved in high-impact decisions.



AI should support human judgment, not quietly replace it.

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A PRACTICAL STARTING POINT

Use AI with curiosity and judgment

AI performs tasks associated with human intelligence.

NLP processes language; computer vision analyzes visual information.

Machine learning and deep learning enable many capabilities.

Multimodal AI connects different kinds of information.

Responsible use requires accuracy, fairness, privacy, security, and accountability.

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